

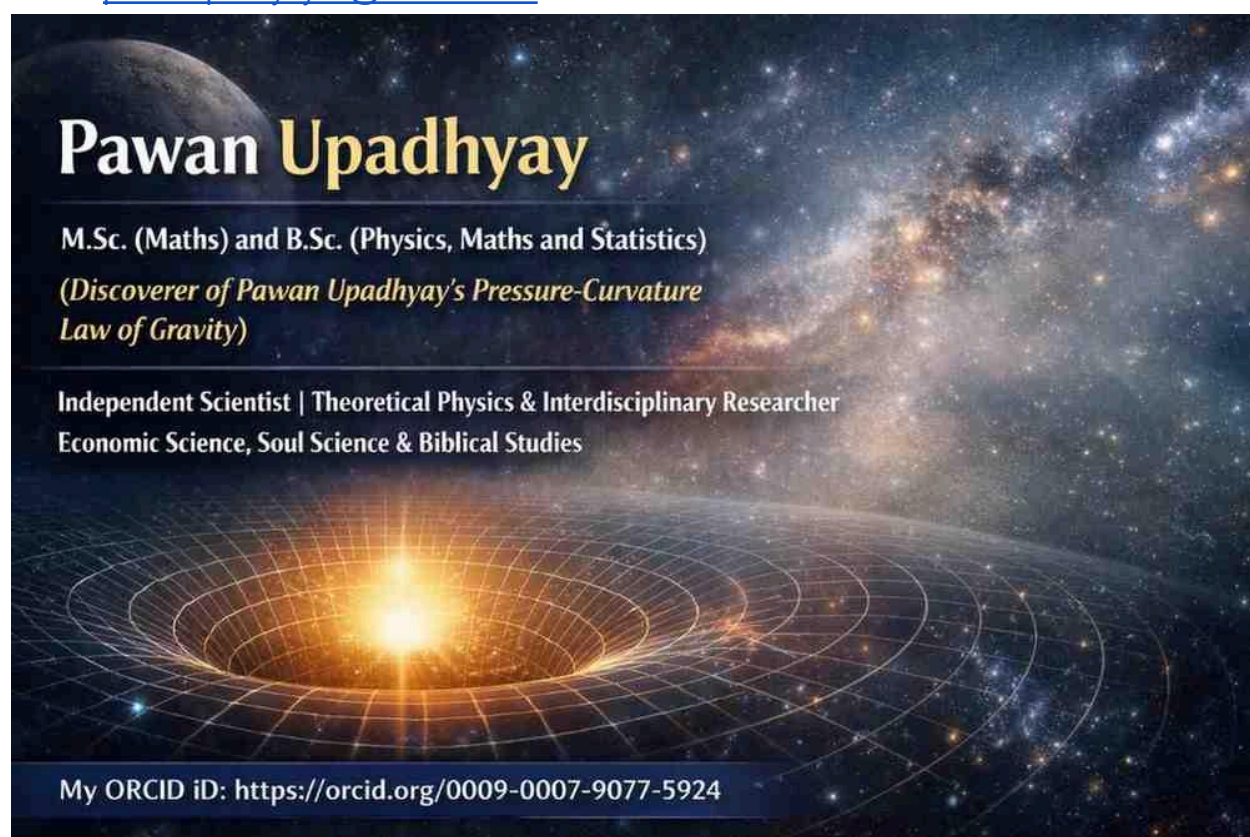
Observational Status of Four-Dimensional Spacetime in Weak-Field and Strong-Field Gravity

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Abstract

The dimensionality of spacetime is a fundamental question in gravitational physics. While numerous theoretical models propose the existence of extra spatial dimensions, all current experimental and observational evidence is consistent with a four-dimensional spacetime consisting of three spatial dimensions and one temporal dimension. This paper reviews the observational status of spacetime dimensionality in both weak-field and strong-field gravitational regimes.

[Abstract] [Expanded]

The observed universe is accurately described by a four-dimensional spacetime consisting of three spatial dimensions and one temporal dimension. Although numerous theoretical frameworks propose additional spacetime dimensions, no direct experimental or observational evidence has yet confirmed their physical existence. Classical gravitational tests within the Solar System, together with modern observations of black holes, neutron stars, and gravitational waves, remain fully consistent with the four-dimensional description provided by General Relativity.

Within this experimentally supported spacetime framework, the Pressure–Curvature (PPC) theory introduces a local equation of state relating the gravitational pressure field to the local energy density field,

$$P_g = \omega E_d,$$

where P_g denotes the gravitational pressure, E_d represents the local energy density, and ω is a dimensionless equation-of-state parameter that may vary throughout spacetime. The theory naturally extends from weak gravitational fields to arbitrary curved spacetime by expressing all physical quantities as spacetime-dependent scalar fields. The strong-field limit corresponds to $\omega \rightarrow 1$, yielding $P_g \approx E_d$.

This paper reviews the current observational status of four-dimensional spacetime and presents the covariant formulation of the PPC equation of state applicable to both weak-field and strong-field gravitational regimes.

Keywords: Four-dimensional spacetime, gravitational pressure, energy density, equation of state, General Relativity, weak-field gravity, strong-field gravity, Pressure–Curvature theory.

1. Introduction

Modern gravitational physics is based on the assumption that spacetime possesses four observable dimensions: three spatial dimensions and one temporal dimension. This framework successfully explains a wide range of physical phenomena, from planetary motion to black holes and gravitational waves.

Several theoretical extensions of gravity introduce additional spacetime dimensions. However, these extra dimensions remain hypothetical and have not been experimentally confirmed.

1. Introduction [Expanded]

One of the fundamental questions in gravitational physics concerns the dimensionality of physical spacetime. Since the development of General Relativity, gravity has been successfully described as the curvature of a four-dimensional spacetime consisting of three spatial dimensions and one temporal dimension. This framework has consistently explained a wide range of gravitational phenomena, from planetary motion within the Solar System to the dynamics of compact astrophysical objects.

Several modern theories, including higher-dimensional gravity and string-inspired models, predict the existence of additional spatial dimensions. Although these theories possess significant mathematical interest, no direct experimental observation has confirmed the existence of extra dimensions. Consequently, all presently verified gravitational phenomena remain accurately described within four-dimensional spacetime.

The Pawan Upadhyay's Pressure–Curvature (PPC) theory is formulated entirely within this experimentally supported four-dimensional framework. Rather than introducing additional dimensions, the theory establishes a local relationship between the gravitational pressure field and the local energy density field through a spacetime-dependent equation of state.

The fundamental PPC equation is

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu),$$

where

- P_g is the gravitational pressure field,
- E_d is the local energy density field,
- ω is the dimensionless equation-of-state parameter,
- $x^\mu = (ct, x, y, z)$ denotes the four-dimensional spacetime coordinates.

Because all quantities are defined locally, the equation remains applicable throughout curved spacetime and naturally extends from weak-field gravity to strong-field gravitational environments.

2. Weak-Field Limit

The weak-field limit applies where gravitational fields are relatively small and spacetime curvature is weak, such as within the Solar System.

Experimental tests in this regime include:

- Planetary orbital motion
- Deflection of light
- Gravitational redshift
- Time dilation
- Shapiro time delay

All of these observations are accurately described by a four-dimensional spacetime (3+1 dimensions). No experimental evidence in the weak-field regime requires the existence of additional spacetime dimensions.

3. Strong-Field Limit

The strong-field limit applies near extremely compact astrophysical objects where spacetime curvature becomes very large.

Examples include:

- Black holes
- Neutron stars
- Binary compact-object mergers
- Gravitational-wave sources

Observations of these systems remain consistent with four-dimensional spacetime as described by General Relativity. Current strong-field observations do not provide experimental confirmation of additional spacetime dimensions.

4. Higher-Dimensional Theories

Several theoretical frameworks propose extra dimensions, including higher-dimensional gravity models and string-inspired theories. These models remain mathematically interesting but currently lack direct observational confirmation.

Consequently, additional spacetime dimensions should presently be regarded as theoretical possibilities rather than experimentally established physical realities.

5. Conclusion

Current observations in both weak-field and strong-field gravitational regimes are consistent with a four-dimensional spacetime composed of three spatial dimensions and one temporal dimension.

To date:

- Weak-field observations support four-dimensional spacetime.
- Strong-field observations are also consistent with four-dimensional spacetime.
- No direct experimental evidence has confirmed the existence of additional spacetime dimensions.

Therefore, four-dimensional spacetime remains the only experimentally supported description of the observed universe.

Keywords

Four-dimensional spacetime, weak-field gravity, strong-field gravity, General Relativity, spacetime dimensions, black holes, gravitational waves, observational gravity.

[Expanded Research Paper]

2. Observational Status of Four-Dimensional Spacetime

Experimental observations consistently demonstrate that the physical universe possesses three observable spatial dimensions. Spatial position is described by the Cartesian coordinates

$$\mathbf{r} = (x, y, z),$$

while the velocity of a particle is expressed as

$$\mathbf{v} = (v_x, v_y, v_z),$$

with magnitude

$$v^2 = v_x^2 + v_y^2 + v_z^2.$$

These relations accurately describe particle motion in ordinary three-dimensional Euclidean space under weak gravitational conditions.

Physical phenomena also evolve with time. Consequently, the observed universe is described by four-dimensional spacetime, whose coordinates are

$$(x, y, z, t),$$

or equivalently,

$$x^\mu = (ct, x, y, z),$$

where c is the speed of light and t denotes the temporal coordinate.

This four-dimensional formulation provides the mathematical foundation for General Relativity and serves as the spacetime framework of the PPC theory.

Within this framework, the PPC equation of state is expressed as

$$P_g(x^\mu) = \omega(x^\mu)E_d(x^\mu),$$

where both the gravitational pressure and the energy density are treated as local scalar fields that may vary throughout spacetime.

3. Weak-Field Formulation of the PPC Equation of State

The weak-field regime corresponds to gravitational systems in which the spacetime curvature is sufficiently small that gravitational effects may be treated as perturbations about flat Minkowski spacetime. This approximation accurately describes most gravitational phenomena observed within the Solar System and many astrophysical environments where the gravitational potential satisfies

$$\left| \frac{\Phi}{c^2} \right| \ll 1,$$

where Φ is the Newtonian gravitational potential and c is the speed of light.

Within the Pressure–Curvature (PPC) theory, the gravitational pressure is postulated to be proportional to the local energy density through the equation of state

$$P_g = \omega E_d,$$

where

- P_g is the gravitational pressure,
- E_d is the local energy density,
- ω is the dimensionless equation-of-state parameter.

In general, each of these quantities may vary with both spatial position and time.

Consequently, the weak-field equation assumes the spacetime-dependent form

$$P_g(x, y, z, t) = \omega(x, y, z, t) E_d(x, y, z, t),$$

The quantity E_d is **not restricted to dark energy density**. Instead, it denotes the total local energy density of the physical system under consideration. Depending on the physical context, E_d may represent the energy density associated with matter, radiation, vacuum energy, relativistic fluids, or any other form of energy contributing to the local gravitational field.

In general, all three quantities may vary with both spatial position and time. Consequently, the PPC equation of state assumes the spacetime-dependent form

$$P_g(x, y, z, t) = \omega(x, y, z, t) E_d(x, y, z, t),$$

or equivalently,

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu),$$

where

$$x^\mu = (ct, x, y, z)$$

denotes the four-dimensional spacetime coordinates.

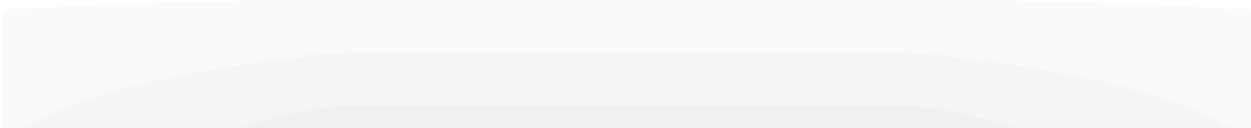
Accordingly, the gravitational pressure field, the equation-of-state parameter, and the energy density are all treated as local scalar fields defined over the observed four-dimensional spacetime. The PPC equation is therefore a local relation that remains valid at every spacetime event.

When the equation-of-state parameter is constant,

$$\omega = \text{constant},$$

the above equation reduces to a special case of the more general spacetime-dependent formulation. The PPC framework therefore accommodates both constant and variable equations of state within a unified mathematical description.

The weak-field formulation establishes the fundamental equation of the PPC theory and provides the basis for its covariant extension to arbitrary curved four-dimensional spacetime.



4. Spacetime Dependence of the PPC Equation of State

A distinguishing feature of the PPC theory is that the equation of state is formulated as a **local spacetime relation** rather than as a global algebraic equation.

Accordingly,

$$P_g = P_g(x^\mu),$$

$$E_d = E_d(x^\mu),$$

and

$$\omega = \omega(x^\mu),$$

are all scalar fields defined over four-dimensional spacetime.

The fundamental PPC equation therefore becomes

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu).$$

This relation states that the gravitational pressure at any spacetime event is determined by the local energy density and the corresponding local value of the dimensionless equation-of-state parameter.

Unlike conventional models that assume a universal constant value of ω , the PPC theory allows the equation-of-state parameter to evolve continuously throughout spacetime. Different physical environments may therefore possess different local equations of state while remaining governed by the same fundamental relation.

The spacetime dependence of ω provides a unified mathematical framework capable of describing weak gravitational fields, relativistic systems, and strong-field gravitational environments without changing the basic structure of the PPC equation. Consequently, constant values of ω represent special cases of a more general spacetime-dependent theory rather than independent physical models.

This local formulation forms the foundation for the covariant extension of the PPC equation of state to arbitrary curved spacetime, which is developed in the following section.

4. Strong-Field Extension of the Pressure–Curvature (PPC) Equation of State

The weak-field formulation of the Pressure–Curvature (PPC) theory is expressed by the local equation of state

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu),$$

where P_g is the gravitational pressure, E_d is the local energy density, and ω is the dimensionless equation-of-state parameter.

To extend the PPC theory to strong gravitational fields, the equation is formulated on a four-dimensional Lorentzian spacetime manifold $(M, g_{\mu\nu})$. The geometry of spacetime is described by the metric tensor $g_{\mu\nu}$, with the invariant spacetime interval

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu,$$

where

$$x^\mu = (ct, x, y, z), \quad \mu = 0, 1, 2, 3.$$

Within this curved spacetime, the PPC equation of state retains its local covariant form,

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu), \quad x^\mu \in (M, g_{\mu\nu}).$$

Here,

- $P_g(x^\mu)$ is the local gravitational pressure field,
- $E_d(x^\mu)$ is the total local energy density field,
- $\omega(x^\mu)$ is the local dimensionless equation-of-state parameter.

The scalar fields P_g , E_d , and ω are defined on the spacetime manifold $(M, g_{\mu\nu})$ and may vary continuously from one spacetime event to another. Consequently, the PPC equation is locally valid throughout both weak-field and strong-field gravitational regimes.

A fundamental postulate of the PPC theory is that physically realizable energy density and gravitational pressure remain finite throughout spacetime. Accordingly,

$$0 \leq E_d(x^\mu) \leq E_{\text{max}},$$

and

$$0 \leq P_g(x^\mu) \leq P_{g,\text{max}}.$$

As the gravitational field approaches its strongest physically attainable regime, such as in the vicinity of a black hole, the PPC theory predicts

$$E_d(x^\mu) \rightarrow E_{\text{max}},$$

and

$$P_g(x^\mu) \rightarrow P_{g,\text{max}},$$

rather than diverging to infinity. Thus, the theory postulates finite upper bounds on the local energy density and gravitational pressure, excluding physical infinities within its framework.

The strong-field extension therefore preserves the same fundamental equation of state as the weak-field theory while embedding it in the curved spacetime geometry of General Relativity. This provides a unified, generally covariant description of gravitational pressure and local energy density across the full range of gravitational fields.

Special Cases of the Equation-of-State Parameter

The general PPC equation of state is

$$P_g(x, y, z, t) = \omega(x, y, z, t) E_d(x, y, z, t),$$

where the equation-of-state parameter ω is a dimensionless function of spacetime.

Particular constant values of ω correspond to special cases of the general formulation:

- $\omega \approx 0$: Pressureless matter (dust),

$$P_g \approx 0.$$

- $\omega \approx \frac{1}{3}$: Radiation or relativistic fluid,

$$P_g \approx \frac{1}{3} E_d.$$

- $\omega \approx 1$: Stiff fluid,

$$P_g \approx E_d.$$

- $\omega \approx -1$: Vacuum energy (cosmological constant),

$$P_g \approx -E_d.$$

Thus, constant numerical values of ω describe specific physical regimes, whereas the PPC theory allows the more general spacetime-dependent relation

$$\omega = \omega(x, y, z, t),$$

which permits transitions between different equation-of-state behaviors across spacetime. This makes the constant values $\omega \approx 0, \frac{1}{3}, 1$, and -1 special cases of the general PPC framework.

4.1 Bounded Energy Density and Gravitational Pressure

A fundamental postulate of the Pressure–Curvature (PPC) theory is that physically realizable energy density and gravitational pressure remain finite throughout the four-dimensional spacetime manifold $(M, g_{\mu\nu})$. Accordingly, the theory excludes divergent energy densities and infinite gravitational pressures.

The local energy density satisfies

$$0 \leq E_d(x^\mu) \leq E_{\max},$$

where $E_{\max}$ denotes the maximum physically attainable local energy density.

Similarly, the gravitational pressure satisfies

$$0 \leq P_g(x^\mu) \leq P_{g,\max},$$

where $P_{g,\max}$ denotes the maximum physically attainable gravitational pressure.

These upper bounds are fundamental postulates of the PPC theory and define the physically admissible range of the local energy density and gravitational pressure throughout spacetime.

Strong-Field Limit of the PPC Equation of State

The PPC theory postulates that, in the strong-field limit, the equation-of-state parameter approaches unity,

$$\omega \rightarrow 1.$$

Consequently, the general PPC equation of state,

$$P_g = \omega E_d,$$

reduces to

$$P_g \approx E_d.$$

4.2 Strong-Field Limit of the PPC Equation of State

The fundamental equation of state of the Pressure–Curvature (PPC) theory is

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu), \quad x^\mu \in M.$$

In the strong-field regime, the PPC theory postulates that the local equation-of-state parameter approaches unity,

$$\omega(x^\mu) \rightarrow 1.$$

Consequently, the PPC equation of state approaches

$$P_g(x^\mu) \rightarrow E_d(x^\mu).$$

Furthermore, the PPC theory assumes that the local energy density remains finite and approaches a maximum physically attainable value,

$$E_d(x^\mu) \rightarrow E_{\max}.$$

Therefore, in the strong-field limit,

$$P_g(x^\mu) \rightarrow P_{g,\max} = E_{\max}.$$

Thus, the strong-field limit of the PPC theory is characterized by

$$\omega \rightarrow 1, \quad E_d \rightarrow E_{\max}, \quad P_g \rightarrow P_{g,\max} = E_{\max}.$$

Accordingly, the PPC theory predicts that, even in the strongest physically realizable gravitational fields, the gravitational pressure and local energy density remain finite and asymptotically approach their respective maximum values. This bounded strong-field behavior distinguishes the PPC framework from theories that permit divergent physical quantities.

When extended to curved four-dimensional spacetime, the strong-field equation is written as

$$P_g(x^\mu) = E_d(x^\mu),$$

where $x^\mu = (ct, x, y, z)$ denotes the spacetime coordinates. The fields P_g and E_d are defined locally on the curved spacetime described by the metric tensor $g_{\mu\nu}$.

Note. Maximum Gravitational Pressure in the PPC Theory

The Pressure–Curvature (PPC) equation of state is

$$P_g = \omega E_d,$$

where ω is the dimensionless equation-of-state parameter.

If the equation-of-state parameter satisfies

$$0 \leq \omega(x^\mu) \leq 1,$$

then the maximum gravitational pressure satisfies

$$P_{g,\max} = \omega_{\max} E_{\max},$$

where E_{max} is the maximum physically attainable local energy density and $P_{g,\text{max}}$ is the corresponding maximum gravitational pressure predicted by the PPC theory.

In the strong-field limit, the PPC theory postulates

$$\omega(x^\mu) \rightarrow 1.$$

Consequently,

$$\omega_{\text{max}} = 1,$$

and the limiting equation becomes

$$P_{g,\text{max}} = E_{\text{max}}.$$

Thus, in the strongest physically realizable gravitational regime, the PPC theory predicts that the maximum gravitational pressure equals the maximum local energy density. This relation follows directly from the PPC equation of state together with the strong-field limit $\omega \rightarrow 1$.

Note 1. Generality of the Energy Density

In the Pressure–Curvature (PPC) theory, E_d denotes the **total local energy density** and is **not restricted to dark energy density**.

Depending on the physical system under consideration, E_d may represent the energy density of ordinary matter, radiation, relativistic fluids, vacuum energy, or any other form of energy contributing to the local gravitational field. Consequently, the PPC equation of state is formulated as a general relation applicable to all physical energy distributions.

Note 2. Bounded Energy Density

A fundamental postulate of the PPC theory is that the local energy density remains finite throughout spacetime. Accordingly,

$$0 \leq E_d(x^\mu) \leq E_{\text{max}},$$

where E_{max} denotes the maximum physically attainable local energy density predicted by the PPC theory. Thus, the theory excludes divergent or infinite energy densities.

Note 3. Bounded Gravitational Pressure

The PPC theory further postulates that the gravitational pressure is also bounded.

Therefore,

$$0 \leq P_g(x^\mu) \leq P_{g,\max},$$

where $P_{g,\max}$ is the maximum physically attainable gravitational pressure. This boundedness follows from the fundamental PPC equation of state together with the finite nature of the local energy density.

Note 4. Strong-Field Limit

As the gravitational field approaches the strongest physically realizable regime, such as in the vicinity of a black hole, the PPC theory predicts that the local energy density and gravitational pressure asymptotically approach finite maximum values:

$$E_d(x^\mu) \rightarrow E_{\text{max}},$$

$$P_g(x^\mu) \rightarrow P_{g,\text{max}}.$$

Accordingly, the PPC theory predicts finite upper bounds on these physical quantities rather than physical infinities.

Note 5. Physical Interpretation

The bounded nature of E_d and P_g is a **fundamental postulate of the PPC theory**. It is introduced as a defining assumption of the theory and is not derived from General Relativity. If supported by future theoretical developments or observational evidence, this postulate would provide a framework in which physically measurable quantities remain finite even in the strongest gravitational fields.

Important Notes:

To extend the PPC theory to strong gravitational fields, the equation is formulated on a four-dimensional Lorentzian spacetime manifold. Let

$$(M, g_{\mu\nu})$$

be a four-dimensional Lorentzian spacetime manifold, where M denotes the spacetime manifold and $g_{\mu\nu}$ is the metric tensor. The geometry of spacetime is described by the invariant spacetime interval

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu,$$

where

$$x^\mu = (ct, x, y, z), \quad \mu = 0, 1, 2, 3.$$

The fields P_g , E_d , and ω are smooth scalar fields defined on the manifold M .

Accordingly, the Pressure–Curvature (PPC) equation of state is

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu), \quad x^\mu \in M.$$

Since P_g , E_d , and ω are scalar fields, the PPC equation retains the same mathematical form under arbitrary smooth coordinate transformations and is therefore generally covariant. Consequently, the equation remains valid throughout both weak-field and strong-field gravitational regimes.

The quantity E_d denotes the **total local energy density** and is not restricted to dark energy. Depending on the physical system under consideration, it may represent the energy density of ordinary matter, radiation, relativistic fluids, vacuum energy, or any other form of energy contributing to the local gravitational field.

A fundamental postulate of the PPC theory is that physically realizable energy density and gravitational pressure remain finite throughout spacetime. Accordingly,

$$0 \leq E_d(x^\mu) \leq E_{\text{max}},$$

and

$$0 \leq P_g(x^\mu) \leq P_{g,\text{max}}.$$

As the gravitational field approaches the strongest physically realizable regime, such as in the vicinity of a black hole, the PPC theory predicts

$$E_d(x^\mu) \rightarrow E_{\text{max}},$$

and

$$P_g(x^\mu) \rightarrow P_{g,\text{max}},$$

rather than diverging to infinity. Thus, the PPC theory postulates finite upper bounds on the local energy density and gravitational pressure, excluding physical infinities within its theoretical framework.

The strong-field extension therefore preserves the same fundamental equation of state as the weak-field formulation while embedding it within the curved spacetime geometry of General Relativity. This provides a unified, generally covariant description of the relationship between gravitational pressure and local energy density across the full range of gravitational fields.

Note 1. Total Local Energy Density

In the Pressure–Curvature (PPC) theory, E_d denotes the **total local energy density** rather than exclusively the dark energy density.

Accordingly, E_d may include contributions from ordinary matter, dark matter, radiation, vacuum energy, relativistic fluids, and any other form of energy present at a given spacetime event. The PPC equation of state is therefore formulated as a general relation applicable to all physical energy distributions.

Note 2. Cosmological Energy Composition

According to the current standard cosmological model (Λ CDM), the total mass-energy content of the present universe is approximately composed of:

- Dark energy: **69%**
- Dark matter: **26%**
- Ordinary (baryonic) matter: **5%**

Radiation and neutrinos contribute only a very small fraction of the present cosmic energy budget. Consequently, the total local energy density in the PPC theory may be expressed schematically as

$$E_d = E_{\text{ordinary}} + E_{\text{dark matter}} + E_{\text{dark energy}} + E_{\text{radiation}} + \cdots ,$$

where the ellipsis denotes any additional local energy contributions.

$$E_d = E_{\text{ordinary}} + E_{\text{dark matter}} + E_{\text{dark energy}} + E_{\text{radiation}} + \cdots ,$$

where the ellipsis denotes any additional local energy contributions.

Note 3. Local Nature of the PPC Equation

The quantity $E_d(x^\mu)$ represents the **local** energy density at the spacetime event x^μ . Therefore, the PPC equation

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu)$$

relates the local gravitational pressure to the total local energy density rather than to the average energy composition of the universe. The cosmological percentages given above describe the large-scale energy budget of the universe and should not be interpreted as the energy composition at every spacetime point.

Note. Special Cases of the Equation-of-State Parameter

The Pressure–Curvature (PPC) equation of state is

$$P_g = \omega E_d,$$

where ω is the dimensionless equation-of-state parameter. Particular constant values of ω correspond to well-known physical regimes:

Equation-of-State Parameter

$$\omega \approx 0$$

$$\omega \approx \frac{1}{3}$$

$$\omega \approx 1$$

$$\omega \approx -1$$

Physical Interpretation	PPC Equation
Pressureless matter (dust)	$P_g \approx 0$
Radiation or relativistic fluid	$P_g \approx \frac{1}{3} E_d$
Stiff fluid (strong-field limit in the PPC theory)	$P_g \approx E_d$
Vacuum energy (cosmological constant)	$P_g \approx -E_d$

In the general PPC theory, the equation-of-state parameter is not required to be constant. Instead,

$$\omega = \omega(x^\mu),$$

allowing it to vary continuously throughout four-dimensional spacetime. The constant values

$$\omega \approx 0, \quad \omega \approx \frac{1}{3}, \quad \omega \approx 1, \quad \omega \approx -1$$

therefore represent special cases of the more general spacetime-dependent PPC equation of state

$$P_g(x^\mu) = \omega(x^\mu) E_d(x^\mu), \quad x^\mu \in M.$$

This formulation provides a unified framework capable of describing different physical regimes through a single spacetime-dependent equation of state.

Interpretation of E_d for Different Values of the Equation-of-State Parameter

In the PPC theory, E_d always denotes the **total local energy density**. However, the dominant physical contribution to E_d depends on the local value of the equation-of-state parameter ω .

Dominant Contribution to E_d	PPC Equation
Ordinary (pressureless) matter (dust)	$P_g \approx 0$
Radiation or relativistic particles	$P_g \approx \frac{1}{3} E_d$
Vacuum (dark) energy	$P_g \approx -E_d$
Maximum local energy density in the PPC strong-field limit	$P_g \approx E_d$

Note

The definition of E_d does **not** change with ω . It always represents the **total local energy density**. The value of ω identifies the physical regime in which a particular form of energy dominates the local energy density. Thus,

- For $\omega \approx 0$, E_d is dominated by ordinary matter.
- For $\omega \approx \frac{1}{3}$, E_d is dominated by radiation.
- For $\omega \approx -1$, E_d is dominated by vacuum (dark) energy.
- For $\omega \approx 1$, the PPC theory associates E_d with the strongest physically realizable gravitational regime, where the local energy density approaches its finite maximum value E_{max} .

This presentation keeps the notation internally consistent: E_d **always means total local energy density**, while ω characterizes which component is dominant in a given physical situation.